

Lecture 6: GLM Estimation — Maximum Likelihood

MATH3701 Statistical Modelling

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Course notes: Chapter 4, Sections 4.1–4.2

The Estimation Problem

Given:

- Observations $y_1, \dots, y_n$
- Explanatory variables $\mathbf{x}_1, \dots, \mathbf{x}_n$
- A GLM specification (distribution, link)

Goal: Estimate the parameter vector $\boldsymbol{\beta} = (\beta_1, \dots, \beta_p)'$

Method: Maximum likelihood estimation (MLE)

$$\hat{\boldsymbol{\beta}} = \arg \max_{\boldsymbol{\beta}} L(\boldsymbol{\beta}; \mathbf{y})$$

The Log-Likelihood Function

For independent observations from exponential family:

$$L(\boldsymbol{\beta}; \mathbf{y}, \phi) = \prod_{i=1}^n f(y_i; \theta_i, \phi)$$

Log-likelihood:

$$\ell(\boldsymbol{\beta}; \mathbf{y}, \phi) = \sum_{i=1}^n \left\{ \frac{y_i \theta_i - b(\theta_i)}{\phi} + c(y_i, \phi) \right\}$$

Key: θ_i depends on $\boldsymbol{\beta}$ through:

$$\theta_i \leftarrow \mu_i = g^{-1}(\eta_i) = g^{-1}(\mathbf{x}_i' \boldsymbol{\beta})$$

The Simple Case: i.i.d. Observations

If all $\theta_i = \theta$ (same for all observations):

$$\ell(\theta; \mathbf{y}, \phi) = \frac{n(\bar{y}\theta - b(\theta))}{\phi} + \text{const}$$

Score equation:

$$\frac{\partial \ell}{\partial \theta} = \frac{n(\bar{y} - b'(\theta))}{\phi} = 0$$

Solution:

$$\boxed{b'(\hat{\theta}) = \bar{y}}$$

Since $\mathbb{E}[Y] = b'(\theta)$, the MLE satisfies $\hat{\mu} = \bar{y}$.

Example: Poisson i.i.d. Case

For $Y_1, \dots, Y_n \stackrel{\text{iid}}{\sim} \text{Poisson}(\lambda)$:

- $b(\theta) = e^\theta$, so $b'(\theta) = e^\theta$
- Score equation: $e^{\hat{\theta}} = \bar{y}$
- Solution: $\hat{\theta} = \log \bar{y}$
- Therefore: $\hat{\lambda} = \bar{y}$

The MLE is the sample mean!

This makes intuitive sense: best estimate of the rate is the observed average.

Directed reading: Chapter 4, Section 4.1.2

The uncertainty on the estimate $\hat{\theta}$ can be shown to be:

$$\text{Var}(\hat{\theta}) \simeq \frac{\phi}{nb''(\theta)}$$

Thus we see that the first two derivatives of $b(\theta)$ play a key role in inference.

The General Case: Non-identical θ_i

When θ_i varies with explanatory variables:

$$\ell(\boldsymbol{\beta}; \mathbf{y}, \phi) = \sum_{i=1}^n \frac{y_i \theta_i(\boldsymbol{\beta}) - b(\theta_i(\boldsymbol{\beta}))}{\phi} + \text{const}$$

Problem: There is no closed-form solution in general.

Exception: Normal linear model has

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$$

For other GLMs: Need iterative methods (Newton-Raphson, Fisher Scoring).

The Score Function

Definition: The score function is the gradient of the log-likelihood:

$$\mathbf{U}(\boldsymbol{\beta}) = \frac{\partial \ell}{\partial \boldsymbol{\beta}} = \begin{pmatrix} \frac{\partial \ell}{\partial \beta_1} \\ \vdots \\ \frac{\partial \ell}{\partial \beta_p} \end{pmatrix}$$

Properties:

- $\mathbb{E}[\mathbf{U}(\boldsymbol{\beta})] = \mathbf{0}$ at true parameter value
- MLE satisfies $\mathbf{U}(\hat{\boldsymbol{\beta}}) = \mathbf{0}$

Observed Fisher information:

$$\mathbf{I}(\boldsymbol{\beta}) = -\frac{\partial^2 \ell}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}'} \quad (p \times p \text{ matrix})$$

Expected Fisher information:

$$\mathbf{J}(\boldsymbol{\beta}) = \mathbb{E}[\mathbf{I}(\boldsymbol{\beta})] = \mathbb{E} \left[-\frac{\partial^2 \ell}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}'} \right]$$

Key result:

$$\mathbf{J}(\boldsymbol{\beta}) = \mathbb{E}[\mathbf{U}(\boldsymbol{\beta})\mathbf{U}(\boldsymbol{\beta})'] = \text{Var}[\mathbf{U}(\boldsymbol{\beta})]$$

Newton-Raphson Method

Iterative algorithm to find $\hat{\beta}$:

Starting from $\beta^{(0)}$, update:

$$\beta^{(t+1)} = \beta^{(t)} + \mathbf{I}^{-1}(\beta^{(t)})\mathbf{U}(\beta^{(t)})$$

Intuition:

- $\mathbf{U}(\beta)$: direction of steepest ascent
- $\mathbf{I}^{-1}(\beta)$: adjusts step size based on curvature

Convergence: Stop when $\|\beta^{(t+1)} - \beta^{(t)}\| < \epsilon$.

Replace observed with expected information:

$$\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} + \mathbf{J}^{-1}(\boldsymbol{\beta}^{(t)})\mathbf{U}(\boldsymbol{\beta}^{(t)})$$

Advantages over Newton-Raphson:

- $\mathbf{J}$ is guaranteed positive definite
- More stable convergence
- Equivalent to iteratively reweighted least squares (IRLS)

For canonical links: Newton-Raphson and Fisher Scoring are identical.

Key points:

- MLEs maximize the log-likelihood $\ell(\boldsymbol{\beta}; \mathbf{y})$
- For i.i.d. case: $\hat{\mu} = \bar{y}$
- General case requires iterative methods
- Newton-Raphson uses observed information
- Fisher Scoring uses expected information (more stable)

Asymptotic Distribution of $\hat{\beta}$

Under regularity conditions, as $n \rightarrow \infty$:

$$\hat{\beta} \sim \mathcal{N}_p(\beta, \mathcal{I}(\beta)^{-1})$$

evaluated at the true β .

- Centred on the true value of β
- Used to estimate standard errors and p-values in the model summary
- Relies on the assumption that n is large.

The Saturated Model

The **saturated model** has one parameter per observation: $\tilde{\mu}_i = y_i$ for each i .

- Achieves the *maximum possible* log-likelihood $\tilde{\ell}$
- Acts as a reference point
- The fitted model M has log-likelihood $\hat{\ell}$

Deviance measures how far M falls short of the saturated model:

$$D = 2\phi(\tilde{\ell} - \hat{\ell})$$

This is the GLM analogue of the residual sum of squares. The deviance is covered in detail in the next lecture.

Key points:

- If data are *iid*, $b(\hat{\theta}) = \hat{\mu} = \bar{y}$
- GLM parameters estimated numerically via Fisher scoring
- Reduces to ordinary LS for the normal model (one step)
- MLEs are asymptotically $\mathcal{N}_p(\boldsymbol{\beta}, \mathcal{J}^{-1})$
- Saturated model = maximum achievable likelihood

Next lecture: Deviance, Pearson/deviance residuals, and fitting GLMs in R.